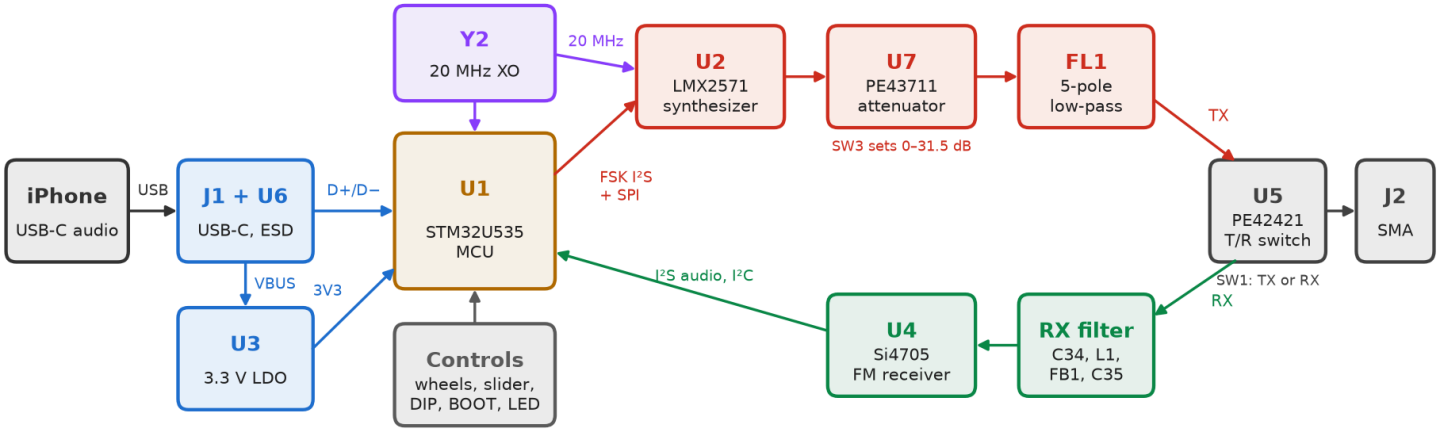
IphoneFM Rev A — design review

MagSafe FM transmitter / receiver for iPhone (USB-C audio). Prepared for an independent sanity check before the JLCPCB order. 26 Oct 2026. Every claim cites its datasheet page. Files and datasheets: github.com/louisflakes/IphoneFM, folder review/rev-a.

1. Overview

What it is. A small board that clips to the back of an iPhone and plugs into its USB-C port. The phone sees a USB audio device. In **transmit** it broadcasts the phone's audio on a chosen FM channel (88.1–107.9 MHz, low-power, FCC Part 15 target). In **receive** it tunes an FM station and streams it back to the phone.

How. The MCU does all the audio work. The radio is two single-chip parts: a TI LMX2571 synthesizer generates the FM carrier directly (the MCU streams the modulation to it as frequency steps), and a Skyworks Si4705 receives. They share one SMA antenna through an RF switch set by a slider.



Board	46.45 × 71.40 mm, 2 layers, 1.6 mm FR-4, 1 oz (JLC0216A stack-up). Ground poured on both layers, 303 vias (123 stitching).
Parts	94 placed (45 types), all on the top side; 5 bare test pads underneath. Fully assembled by JLCPCB.
Power	USB-C 5 V from the phone → AP2112K 3.3 V LDO (600 mA). ~96 mA transmitting, ~62 mA receiving, plus ≤ 15 mA LED. Inside Apple's 100 mA (before USB set-up) / 500 mA limits.
Transmit	88.1–107.9 MHz, direct digital FM ±75 kHz, stereo multiplex built in the MCU. Level: LMX2571 −18…+3 dBm, then a 0–31.5 dB DIP-switch attenuator, then a 5-pole low-pass (harmonics est. ≤ −53 dBc).
Receive	Si4705, I²S digital audio to the MCU, then to the phone as USB audio.
Reference	One 20 MHz XO (±20 ppm) clocks the MCU, the synthesizer and, through an MCU timer, the receiver.
RF line	50 Ω grounded coplanar, 0.92 mm track, 0.20 mm gap. Bottom layer under the RF section is solid ground except two short 3V3 links.
Controls	3 thumbwheels (channel digits), TX/RX slider, 6-bit power DIP switch, BOOT button, RGB LED.
Cost	2 assembled boards (of 5 made) ≈ $190 before shipping (JLC quote, 26 Oct 2026).

Main parts

Ref	Function	Part (maker)	LCSC	$ / board	JLC stock
U1	MCU: USB audio, DSP	STM32U535CEU6 (ST)	C22454224	10.08	0 → global sourcing
U2	TX synthesizer, direct FM	LMX2571NJKR (TI)	C440388	11.32	142
U4	FM receiver, digital audio	Si4705-D60-GM (Skyworks)	C1526114	3.11	48
U7	TX attenuator	PE43711B-Z (pSemi)	C470929	3.91	1,673
U5	T/R antenna switch	PE42421SCAA-Z (pSemi)	C470893	2.92	669
Y2	20 MHz CMOS reference	OT322520MJBA4SL (YXC)	C725987	0.54	4,030
U3	3.3 V LDO	AP2112K-3.3TRG1 (Diodes)	C51118	0.18	10,885
U6, D3–D5	ESD: USB, CC, antenna	USBLC6-2SC6 (ST), TPD1E05U06 (TI), RClamp0521P (Semtech)	C7519, …	0.36	deep
J1 / J2	USB-C socket / SMA jack	TYPE-C-31-M-12 (HRO) / BWSMA-KWE-Z001	C165948 / C496551	0.72	deep
RV1–RV3	Tuning wheels	RK10J11R0A0H (Alps)	C470766	2.17	908

SW1 / SW2 / SW3	Slider / BOOT / DIP	MSK12C02 / EVQ-P7J01P / TDP06H1SBD1 (C&K)	C431540, …	3.63	SW3: global sourcing

Passives: 0402–0805; 16 of the 45 part types are JLC Basic parts. Full BOM, placement file and assembly drawing are in the package.

Fabrication and assembly

- Our own 1-up panel (KiKit): 6.5 mm rails all round, because parts overhang every edge (wheels, SMA barrel, USB-C, switches); 5 mouse-bite tabs; 3 fiducials.
- DRC: 0 errors. One rule exception: J1's two unplated locating holes sit 0.22 mm from its pads (maker's land pattern; JLC minimum 0.20 mm).
- Placement file corrected per part against JLC's own footprints (rotation and origin), and checked in JLC's preview. JLC engineer placement confirmation is on.
- Gerbers read back with an independent viewer and match the board.

Firmware plan (not written yet)

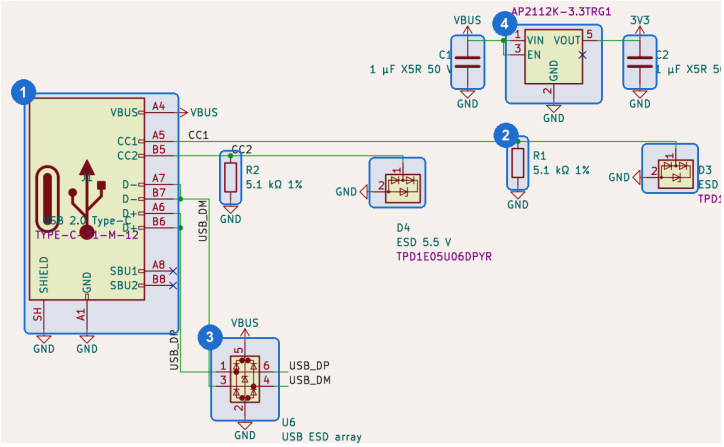
1. **Load:** STM32 ROM USB bootloader (hold BOOT while plugging in); SWD pads as backup.
2. **USB:** Audio Class 2, 48 kHz stereo, asynchronous with feedback (Apple's recommendation).
3. **TX:** MCU builds the FM stereo multiplex (75 µs pre-emphasis, 19 kHz pilot, 38 kHz L−R), corrects the PLL response, interpolates to ~912 kSa/s and streams it to the LMX2571 over SAI in FSK I^2S mode (TI SNAA309). LMX2571 set up over bit-banged MICROWIRE; lock watched on MUXout.
4. **RX:** Si4705 over I^2C; its I^2S audio into the second SAI block, back to the phone.
5. **UI:** wheels on the ADC → channel; slider read for mode (it also switches the RF path directly); LED shows state.

2. Signal chains

Each chain is drawn on the actual schematic; the numbered badges follow the signal. Page references are to the datasheet in the package.

Power and USB

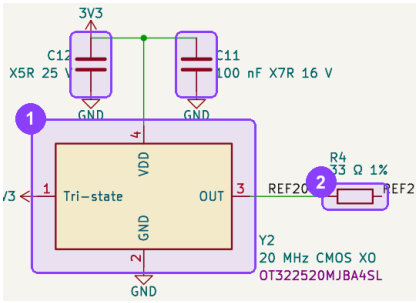
1 J1 USB-C → 2 R1/R2 Rd + D3/D4 ESD → 3 U6 ESD (D+, D−, VBUS) → 4 U3 LDO → 3V3



- 5.1 kΩ Rd on each CC line makes the phone switch on 5 V; no USB-PD needed. USB Type-C R2.5 Table 4-28 p246
- USBLC6 at the connector on D+, D− and VBUS; TPD1E05U06 on each CC. USBLC6-2 p1–2
- AP2112K: 600 mA, stable with ≥ 1 µF; load ≤ ~110 mA. AP2112 p1–2
- Ferrites split the ADC supply (VDDA, FB2) and the receiver supply (FB3) from digital 3V3. DS14217 Fig. 23 p127; Si4705 p16

Reference clock

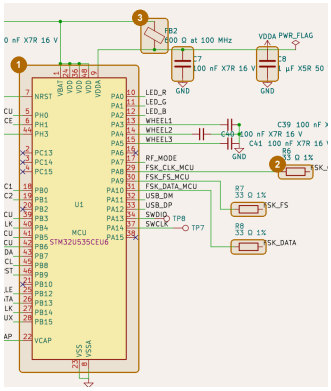
1 Y2 20 MHz XO → U2 OSCin and 2 R4 33 Ω → U1 PH0



- CMOS straight into OSCin, the input TI recommends for best spurs. LMX2571 Table 37 p38, p11
- U1 runs in HSE bypass from the same clock. DS14217 p184–185
- U1 timer makes 32,787 Hz for the Si4705's RCLK (31.1–34.4 kHz allowed). AN332 p37

MCU

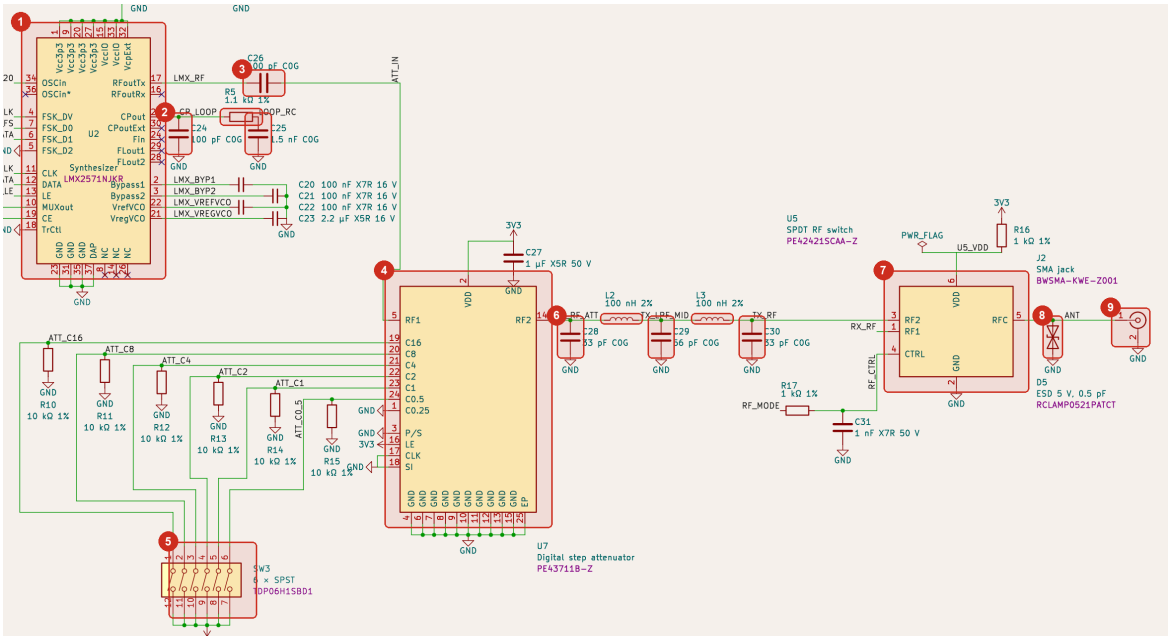
1 U1 STM32U535 → 2 R6–R8 33 Ω FSK I²S · 3 FB2 VDDA



- Bypassing per ST's power scheme; 4.7 µF on VCAP. DS14217 p127
- Pins checked against the alternate-function and bootloader tables. DS14217 Tables 26–28; AN2606 Table 165

Transmit

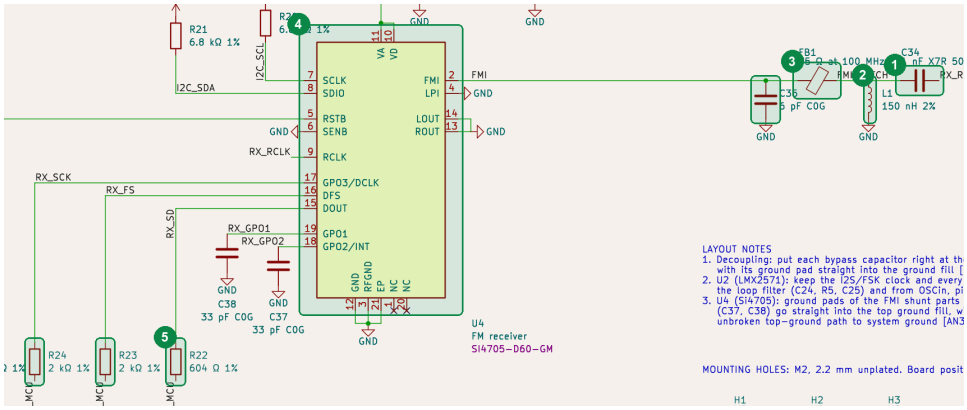
1 U2 LMX2571 → 2 loop filter · 3 C26 DC block → 4 U7 attenuator ← 5 SW3 → 6 FL1 low-pass → 7 U5 switch → 8 D5 ESD → 9 J2



- LMX2571 per TI's typical schematic: 100 nF on every supply pin, 2.2 µF on VregVCO, VcpExt powered. LMX2571 Fig. 62 p44; p53–54
- Loop filter 100 pF / 1.1 kΩ / 1.5 nF → 300–405 kHz loop, ~61° margin; flat to 15 kHz, +0.6 dB at 53 kHz (MCU pre-corrects). Charge-pump current is a firmware setting. LMX2571 p5–6, p35
- Divider ÷56 / ÷48 and two reference settings give ±100 kHz deviation headroom on all 100 US channels. LMX2571 Eq. 4 p35; SNAA309 §6.2
- PE43711 in direct parallel mode from the DIP switch; pull-downs make an open switch read 0. PE43711 p8, Table 4 p6
- FL1: 5-pole 50 Ω low-pass, 23–34 dB at the 2nd harmonic, ≥ 43 dB from 264 MHz. 47 CFR §15.239 / §15.209
- PE42421: ~85 dB isolation keeps the TX carrier off the receiver; the slider drives its CTRL pin directly. PE42421 p3–4

Receive

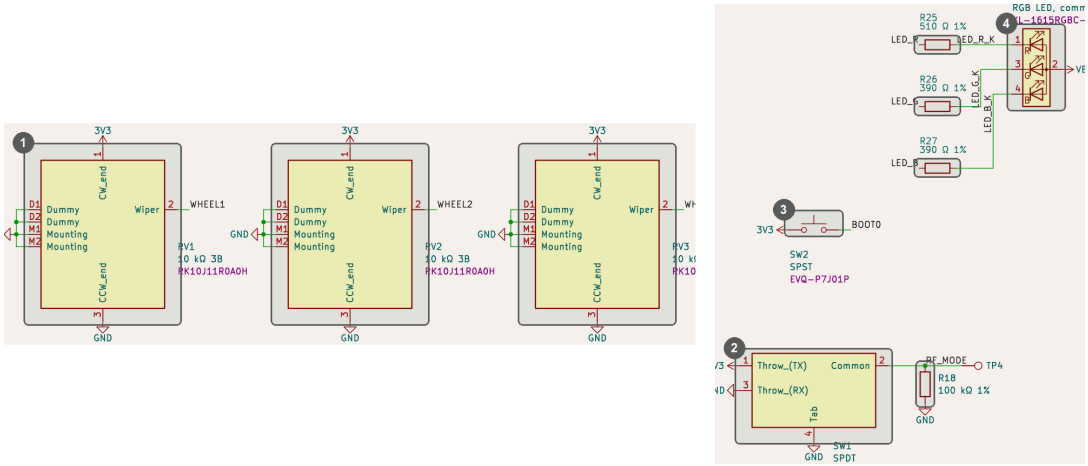
J2 → U5 RF1 → 1 C34 1 nF → 2 L1 150 nH match → 3 FB1 + C35 filter → 4 U4 Si4705 → 5 R22–R24 → U1 SAI



- Front end, emissions filter and I²S series parts as Skyworks specifies. Si4705 p16, Table 12 p18; AN383 p7–9, p37
- U2 is powered down (CE) while receiving, so its carrier can't desensitise U4. LMX2571 p8

Controls

1 RV1–RV3 → ADC · 2 SW1 → RF_MODE · 3 SW2 BOOT0 · 4 D1 LED



- Wheels are 10 kΩ dividers 3V3–GND, read by the ADC; firmware splits each into 10 digits.
- Slider: 3V3 = TX, GND = RX, 100 kΩ holds RX between contacts; it drives U5 and is read by U1. PE42421 p4
- BOOT0 pulled low; pressed at plug-in → ROM USB bootloader. AN2606 p30
- LED anode on VBUS (blue/green need up to 3.4 V), sunk by U1's 5 V-tolerant open-drain pins. XL-1615 p3–4; DS14217 p132

3. What I'd most like checked

1. **Direct FM with the LMX2571.** No published schematic uses it this way; is the loop filter (C24, R5, C25) and the FSK I²S approach sound?
2. **RF layout on 2 layers:** the 50 Ω line, ground stitching and return paths under U2, U5 and U4, and TX-to-RX isolation.
3. **FL1 values** (33 pF, 100 nH, 56 pF, 100 nH, 33 pF) against a square-wave source, and whether its harmonic margin is enough for Part 15.
4. **USB-C sink:** Rd only, no PD controller; current budget against Apple's limits.
5. **Clocking:** one 20 MHz XO for MCU and synthesizer; 32,787 Hz from a timer for the Si4705.
6. **Manufacturing:** the panel and rails, the J1 hole-clearance exception, the JLC placement corrections.
7. Anything that would stop it working on first power-up.

Files

github.com/louisflakes/IphoneFM, folder **review/rev-a**: this PDF, the schematic (PDF), the JLC panel Gerbers, BOM and placement file, the assembly drawing, board pictures, and a README linking the KiCad project and every datasheet cited.